

Common Table Expressions

Finding the average amount paid by the top 5 customers **using CTE**.

```
WITH average_spend_cte AS
(SELECT SUM(amount) AS total_amount_paid
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
'Pingxiang','Sivas','Celaya','So Leopoldo')
AND country IN ('India','China','United States','Japan','Mexico','Brazil',
'Russian Federation','Philippines','Turkey','Indonesia')
GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
ORDER BY SUM(amount) DESC
LIMIT 5)
SELECT AVG(total_amount_paid) AS avg_amount_paid
FROM average_spend_cte
```

The screenshot displays a SQL IDE interface. The top pane, titled 'Query History', contains the SQL query used to find the average amount paid by the top 5 customers. The bottom pane, titled 'Data Output', shows the results of the query. The results table has two columns: 'avg_amount_paid' (numeric) and a single row with the value '107.354000000000000000'. The interface also includes a toolbar with icons for file operations, a status bar at the bottom indicating 'Total rows: 1' and 'Query complete 00:00:02.117', and a 'Showing rows: 1 to 1' indicator.

```
Query History
```

```
1 WITH average_spend_cte AS
2 (SELECT SUM(amount) AS total_amount_paid
3 FROM payment A
4 INNER JOIN customer B ON A.customer_id = B.customer_id
5 INNER JOIN address C ON B.address_id = C.address_id
6 INNER JOIN city D ON C.city_id = D.city_id
7 INNER JOIN country E ON D.country_id = E.country_id
8 WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
9 'Pingxiang','Sivas','Celaya','So Leopoldo')
10 AND country IN ('India','China','United States','Japan','Mexico','Brazil',
11 'Russian Federation','Philippines','Turkey','Indonesia')
12 GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
13 ORDER BY SUM(amount) DESC
14 LIMIT 5)
15 SELECT AVG(total_amount_paid) AS avg_amount_paid
16 FROM average_spend_cte
```

Data Output Messages Notifications

	avg_amount_paid numeric
1	107.354000000000000000

Showing rows: 1 to 1 Page No:

Total rows: 1 Query complete 00:00:02.117

Finding out how many of the top 5 customers I identified in step 1 are based within each country **using CTE**.

```
WITH top_5_customers_cte AS
(SELECT A.customer_id,
B.first_name,
B.last_name,
D.city,
E.country,
SUM(amount) AS total_amount_paid
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
'Pingxiang','Sivas','Celaya','So Leopoldo')
AND country IN ('India','China','United States','Japan','Mexico','Brazil',
'Russian Federation','Philippines','Turkey','Indonesia')
GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
ORDER BY SUM(amount) DESC
LIMIT 5)
SELECT cntry1.country,COUNT(DISTINCT cust1.customer_id) AS all_customer_count,
COUNT(DISTINCT top_5_customers_cte.customer_id) AS top_5_customers
FROM customer AS cust1
INNER JOIN address AS add1 ON cust1.address_id=add1.address_id
INNER JOIN city AS city1 ON add1.city_id=city1.city_id
INNER JOIN country AS cntry1 ON city1.country_id=cntry1.country_id
LEFT JOIN top_5_customers_cte ON cntry1.country=top_5_customers_cte.country
GROUP BY cntry1.country
ORDER BY top_5_customers DESC, all_customer_count DESC
```

Query

Query History

```

1 WITH top_5_customers_cte AS
2 (SELECT A.customer_id,B.first_name,B.last_name,D.city,E.country,
3 SUM(amount) AS total_amount_paid
4 FROM payment A
5 INNER JOIN customer B ON A.customer_id = B.customer_id
6 INNER JOIN address C ON B.address_id = C.address_id
7 INNER JOIN city D ON C.city_id = D.city_id
8 INNER JOIN country E ON D.country_id = E.country_id
9 WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
10 'Pingxiang','Sivas','Celaya','So Leopoldo')
11 AND country IN ('India','China','United States','Japan','Mexico','Brazil',
12 'Russian Federation','Philippines','Turkey','Indonesia')
13 GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
14 ORDER BY SUM(amount) DESC
15 LIMIT 5)
16 SELECT cntry1.country,COUNT(DISTINCT cust1.customer_id) AS all_customer_count,
17 COUNT(DISTINCT top_5_customers_cte.customer_id) AS top_5_customers
18 FROM customer AS cust1
19 INNER JOIN address AS add1 ON cust1.address_id=add1.address_id
20 INNER JOIN city AS city1 ON add1.city_id=city1.city_id
21 INNER JOIN country AS cntry1 ON city1.country_id=cntry1.country_id
22 LEFT JOIN top_5_customers_cte ON cntry1.country=top_5_customers_cte.country
23 GROUP BY cntry1.country
24 ORDER BY top_5_customers DESC, all_customer_count DESC

```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 108

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 Page No: 1

	country character varying (50)	all_customer_count bigint	top_5_customers bigint
1	Mexico	30	2
2	India	60	1
3	United States	36	1
4	Turkey	15	1
5	China	53	0
6	Japan	31	0

Total rows: 108

Query complete 00:00:02.049